Set up a Chromebook for Fledge

An adult account, Linux, VS Code and the ESP32 tools, in about 30 minutes.

This guide assumes you are comfortable with a terminal but have not used Linux on a Chromebook. Google's menus change from time to time, so if a label differs, look for the closest match. GitHub sign-in and Claude are not covered.

Step	You do	Time
1	Add an adult account	5 min
2	Turn on Linux	10 min
3	Run the setup script (mostly waiting)	10–15 min
4	Share the ESP32 with Linux and check it	2 min

Before you start

- A personal Chromebook that supports Linux. Most made since 2019 do.
- About 10 GB of free space for Linux. The ESP32 tools alone use about 8 GB.
- Power and steady Wi-Fi. The script downloads a lot.
- A work or school Chromebook may have Linux turned off by its administrator. If so, only the administrator can change that.

1. Add an adult account

Use a regular Google Account for an adult. Linux is generally not available on a child account supervised with Family Link, so a child's account won't work for this setup. If you already sign in as an adult on this Chromebook, skip to step 2.

1. Sign out of the Chromebook.
2. On the sign-in screen, select **Add person** at the bottom.
3. On **Choose your setup**, pick personal use, then **Next**.
4. Sign in with the adult's Google Account and follow the prompts.

If you see “Sorry, you don't have permission to sign in”, the Chromebook's owner (the first account added to it) has to allow it. They open **Settings**, **Privacy and security**, **Manage other people**, and either add your account to the list or turn off **Limit who can sign in**. Ownership can't be moved to another account.

2. Turn on Linux

This creates a Debian Linux virtual machine that lives inside your account. Other people on the Chromebook won't see it, and it doesn't touch ChromeOS.

1. Click the time in the bottom-right corner, then open **Settings**.
2. Go to **About ChromeOS**, then **Developers**.
3. Next to **Linux development environment**, select **Set up**.
4. Accept the suggested username, and set the disk size to at least **20 GB** if you are asked.
5. Wait for a Terminal window to open. It usually takes about 10 minutes.

3. Run the setup script

In the Terminal window, paste this as one line and press Enter (Ctrl+Shift+V pastes in Terminal):

```
curl -fsSL
"https://raw.githubusercontent.com/cranberry314/Fledge/main/Chrombook%20Setup/setup.txt"
-o setup.txt && bash setup.txt
```

It installs Git, Python, VS Code, arduino-cli, the ESP32 tools and the RF24 radio library, then downloads Fledge to a folder called **Fledge**. The ESP32 tools are the slow part. If it stops or the Wi-Fi drops, run **bash setup.txt** again and it picks up where it left off.

When it prints **All done**, shut Linux down so the new USB permission takes effect: right-click the Terminal icon on the shelf and choose **Shut down Linux**, then open Terminal again. To open the project:

```
code ~/Fledge
```

4. Share the ESP32 with Linux

ChromeOS keeps USB devices away from Linux until you allow it.

1. Plug in the ESP32. If ChromeOS offers to connect it to Linux, accept.
2. If it doesn't, open **Settings**, **About ChromeOS**, **Developers**, **Linux development environment**, **Manage USB devices**, and turn the ESP32 on. It may need sharing again after you unplug it.
3. In Terminal, run the line below. Seeing **/dev/ttyUSB0** or **/dev/ttyACM0** means it worked.

```
ls /dev/ttyUSB* /dev/ttyACM*
```

To build the relay and put it on the ESP32, swap in your port if it differs:

```
cd ~/Fledge/firmware
arduino-cli compile --fqbn esp32:esp32:esp32 Relay
arduino-cli upload --fqbn esp32:esp32:esp32 -p /dev/ttyUSB0 Relay
```

If something goes wrong

What you see	What to do
No **Developers** section or no Linux option	The Chromebook is too old, or an administrator or Family Link has blocked it.
“Linux has only N GB free”	Make the disk bigger in the Linux settings, then run the script again.
Script stops partway	Run **bash setup.txt** again.
No ESP32 in the **ls** line	Try a different cable (some only carry power), then check Manage USB devices.
“Permission denied” on the port	Shut down Linux and open Terminal again.
VS Code or Linux feels slow or closes	Close Chrome tabs. Linux shares the Chromebook's memory, and 8 GB is tight for VS Code plus a big install.